

Alex Lee

Waterloo, Canada | a55lee@uwaterloo.ca | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SKILLS

Languages: Python, Java, C/C++, R, SQL(MySQL, MongoDB), Bash
Data/ML: Pandas, Matplotlib, SciKit, Pytorch, TensorFlow, NumPy, OpenCV
Tools: PowerBI, MS Office (Excel, Word, PowerPoint), Linux, Git, AWS

EXPERIENCE

Machine Learning Engineer

Jan. 2026 – Aug. 2026

City of Edmonton - Future Cities Institute

Edmonton, AB

- Architected and deployed **CityVision**, an open-source real-time traffic control software, achieving **65% cost savings** in city infrastructure operations through automated vehicle detection.
- Optimized a **YOLOv11** vehicle detection model with an automated data pipeline on **GCP** for continuous training, improving model inference efficiency and lifecycle management.
- Engineered a real-time monitoring dashboard using **Shiny R** to provide stakeholders with interactive traffic analytics and live insights for infrastructure decision-making.
- Containerized the entire application stack using **Docker** to ensure a scalable environment and streamlined deployment via **CI/CD pipelines** (GitHub Actions/GCP).
- Collaborated in an **Agile** environment, regularly presenting technical progress and model performance metrics to stakeholders to facilitate data-driven urban policy improvements.

Finance Lead

Sep. 2024 - Apr. 2025

Association of Korean Canadian Scientists and Engineers

Waterloo, ON

- Conveyed accountability and organization skills by analyzing and managing the cash flow of **\$1,000+** annual income and expenses using **Excel**.
- Reduced **10%** of unnecessary expenditures by writing annual financial reports to identify inefficiencies and recommending actionable strategies for better financial management of the organization.

PROJECTS

FBREF Football Data Analysis | [GitHub](#)

- Collected **100,000+** football match data using **Selenium** and organized the data into a **Pandas DataFrame**.
- Identified **progress passing distance** as key predictor of math results, suggesting focused strategies to teams.

Credit Card Fraud Detection | [GitHub](#)

- Developed credit card fraud classification model using **sklearn** and **TensorFlow** with **96%** prediction accuracy.
- Processed and cleaned **200,000+** transaction records, focusing on handling imbalanced financial datasets.

Customer Churn Prediction Model | *Python, SciKit, Pandas, Streamlit* | [GitHub](#)

- Built a **machine learning model** to predict customer churn, using **900+** telecom industry customers' datasets.
- Utilized the model with interactive **Streamlit** web app to improve accessibility and gain actionable insights.
- Identified **central factors**(tenure, monthly charges), helping companies plan their products more effectively.

Academic Management System | *Spring, Javascript, HTML/CSS, MySQL, jQuery, Bootstrap* | [GitHub](#)

- Built a full-stack academic management web app of **20+** pages with **50+** users using **Spring** and **MySQL**.
- Distributed project on **AWS EC2** with **Apache Tomcat**, with optimized database schema and RESTful APIs.
- Implemented **10+** essential features, **CRUD** operations of course postings and assignments, self implemented real-time chat, custom course search filters, and dynamic operations and views with the database.
- Reduced query response time by **30%** using nested SELECT statements to handle grouped ID increments.
- Ensured reliability with **200+** test cases and resolved issues such as multiple PK, time zone mismatches.

EDUCATION

University of Waterloo

Sep. 2023 - Apr. 2028

Bachelor of Mathematics, Major in Data Science

Waterloo, ON

Minors: Combinatorics and Optimization, Management

Relevant Coursework: Data Structures, Algorithms, OOP, Statistics, Linear Models